

Graphite Interlinked Metamaterials (PAMs)

Lattice Architecture, Catenation Logic, Clearance Physics & Simulation Handoff Report

Target Audience: CAE Simulation Engineers & Multi-Body Dynamics Researchers

Subsystem:
graphite.explicit.interlinked

Status: Validated Production Engine (100+ Tests Passing)

HIGH-LEVEL LOGIC SUMMARY: WHAT IS A PAM IN PLAIN ENGLISH?

A standard 3D printed lattice is a *welded truss* where every strut is fused at common vertices. A **Polycatenated Architected Material (PAM)** is fundamentally different: it is true **3D chainmail** made of hundreds to thousands of **independent, unbonded, hollow polyhedral wireframe cages** that are topologically intertwined in space. Every cage floats with a strictly positive physical air gap ($\Delta > 0$) to its neighbors. Because there are zero solid welds between cages, the material exhibits a distinct **kinematic free-play regime**, followed by **geometric contact locking** and **giant energy dissipation via sliding friction**.

1. Foundational Mechanics: Why PAMs Behave Differently than Welded Lattices

When simulating standard explicit lattices (e.g. Kelvin, Octet, Kagome), finite element solvers rely on linear elastic continuum assumptions: struts stretch and bend axially, and deformation is governed by parent material Young's modulus and strut cross-sectional inertia. In PAMs, continuum assumptions fail because global compliance is dominated by **contact mechanics and rigid-body free-play**:

- **Phase I: Kinematic Articulation (Zero Initial Stiffness):** Under small applied loads, cages slide and rotate freely within the clearance envelope without any material strain.
- **Phase II: Geometric Contact Jamming:** As displacement grows, adjacent strut surfaces make contact across multiple coordination directions, forming an interconnected load-bearing network.
- **Phase III: Multi-Body Coulomb Friction Dissipation:** High normal contact forces under cyclic or impact loading cause strut surfaces to slide against one another. Inter-cage friction ($\mu \approx 0.15 - 0.35$) dissipates massive kinetic and acoustic energy without plastic fatigue of the base metal or polymer.

2. How We Construct 3D Interlocking Lattices

HIGH-LEVEL LOGIC SUMMARY: HOW DO CAGES INTERLINK IN 3D WITHOUT COLLIDING?

In 2D, linking chainmail rings is easy. But in 3D, if you place closed cages at cubic lattice sites, how do you prevent them from crashing into each other along all 3 axes at once? Simple shapes like cuboctahedra fail because 3D rotations do not commute—you cannot twist a cage by 45° in X, Y, and Z simultaneously. Our solution uses **Archimedean Truncated Tetrahedra (C-6-TT)**, which have open hexagonal and triangular window cutouts naturally facing all 6 Cartesian neighbor directions. Adjacent cages slide right through these open windows with *zero relative twist* and zero collisions. On diamond networks, we use **Regular Tetrahedra (D-4-TET)** on a bipartite grid with a 60° bond twist, achieving 4-fold corner-through-face catenation.

The fundamental mathematical challenge in 3D catenation is avoiding **rotational frustration**:

- **The Cuboctahedron Paradox:** Setting a standard wireframe cuboctahedron on a simple cubic (pcu) grid causes collisions ($\Delta = 0.0$ mm). Because orthogonal $SO(3)$ rotations do not commute ($[R_x, R_y] \neq 0$), no static 3D rotation matrix can align face windows for clearance along all three Cartesian axes at once.
- **The C-6-TT Breakthrough (Zhou et al., Science 2025):** Truncating a regular tetrahedron at $1/3$ edge length yields 12 vertices and 18 struts. Its 4 hexagonal faces align with $\langle 1, 1, 1 \rangle$ directions, and its 4 triangular corner windows align with Cartesian approach vectors. Neighboring cages penetrate through these windows unrotated, achieving **6-fold coordination ($z = 6$)** with generous positive clearance.

- **The D-4-TET Diamond Architecture:** On diamond-cubic networks (`dia` , coordination $z = 4$), cages are regular tetrahedra (4 vertices, 6 struts). The engine splits the lattice into bipartite Sublattices A and B. Sublattice B is inverted and twisted by 60° relative to Sublattice A along the tetrahedral bond axis, enabling **corner-through-face catenation**.
- **Topological Proof (Gauss Linking Integral):** To verify true multi-body catenation prior to meshing, the engine computes the double Gauss linking path integral:

$$Lk(\gamma_1, \gamma_2) = (1 / 4\pi) \oint \oint [(r_1 - r_2) \cdot (dr_1 \times dr_2)] / |r_1 - r_2|^3 \neq 0$$

A non-zero linking number mathematically guarantees that cages cannot be separated without cutting.

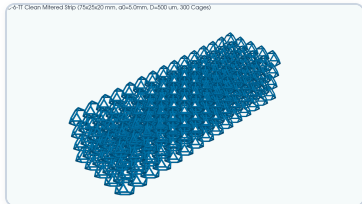


Figure 1: C-6-TT strip (75 × 25 × 20 mm, 300 cages, $\Delta = 157 \mu\text{m}$).

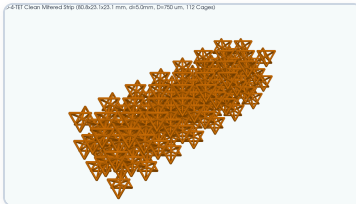


Figure 2: D-4-TET strip (80.8 × 23.1 × 23.1 mm, 112 cages, $\Delta = 427 \mu\text{m}$).

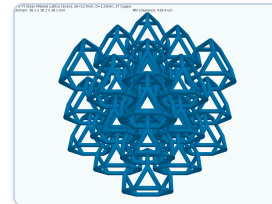


Figure 3: 3 × 3 × 3 C-6-TT cube ($a_0 = 12.7 \text{ mm}$, $\Delta = 418 \mu\text{m}$).

3. The Lattice Generation Engine & Clearance Scaling Physics

HIGH-LEVEL LOGIC SUMMARY: HOW DO WE GENERATE LATTICES AND GUARANTEE THEY NEVER FUSE?

We don't draw every cage from scratch. We define a single prototype cage at the origin and clone it across space using **SE(3) rigid transformation matrices** (position + orientation). This generates 500+ cages in under 1 second. To guarantee cages never touch during 3D printing, we use a **two-tier clearance engine**: a fast spatial bounding-sphere filter (cKDTree) prunes 90% of non-neighbor pairs, and vectorized 3D geometry finds the exact closest approach between all remaining struts. Clearance scales linearly with cell size: $d_{\min} = \kappa \cdot a$. If a simulation or manufacturing engineer specifies a desired strut diameter and clearance, our **closed-form pitch solver** calculates the exact required unit cell pitch in one formula.

3.1 Memory-Efficient SE(3) Rigid Particle Abstraction

Each particle is an independent `InterlinkedParticle` instance governed by an SE(3) matrix: $T = [R \mid t]$, $R \in SO(3)$, $t \in \mathbb{R}^3$ $\implies V_{\text{global}} = V \cdot R^T + t^T$. Vertices and faces are evaluated lazily, allowing Graphite to assemble thousands of kinematic bodies with minimal memory and zero vertex-merging bugs.

3.2 Two-Tier Clearance Verification Pipeline

1. **Tier 1 (Broad Phase):** Evaluates a spatial `scipy.spatial.cKDTree` across cage centroids. Only pairs satisfying $\text{dist}(c_A, c_B) < 2.2 \cdot R_{\text{outer}} + 2r$ are retained (pruning >90% in $O(N \log N)$ time).
2. **Tier 2 (Narrow Phase):** For retained pairs, computes the exact 3D segment Euclidean distance: $d(s_A, s_B) = \min || p_A(t_A) - p_B(t_B) ||$.
3. **Surface Clearance Evaluation:** Physical clearance is $\Delta = d_{\min} - D$. The DfAM guardrail halts execution if $\Delta \leq 0$, preventing accidental solid collisions before exporting meshes.

3.3 Crystallographic Clearance Constants & Closed-Form Pitch Inversion

Because cage geometry scales proportionally with unit cell pitch `a`, the minimum centerline distance satisfies a strict linear scaling law: $d_{\min} = \kappa \cdot a$.

Lattice Topology	Network Symmetry	Characteristic Pitch Metric (a)	Clearance Constant (κ)	Governing Clearance Law
C-6-TT	Simple Cubic (<code>pcu</code>)	Lattice repeat pitch a_0	$\kappa \approx 0.1314$	$\Delta = 0.1314 \cdot a_0 - D$
D-4-TET (Conv)	Diamond Cubic (<code>dia</code>)	Conventional cube size a_{conv}	$\kappa \approx 0.1018$	$\Delta = 0.1018 \cdot a_{\text{conv}} - D$
D-4-TET (Bond)	Diamond Cubic (<code>dia</code>)	Bond length d_{bond}	$\kappa \approx 0.2354$	$\Delta = 0.2354 \cdot d_{\text{bond}} - D$
European 4-in-1	Planar Maille Weave	Ring center pitch a_{ring}	$\kappa \approx 0.2200$	$\Delta = 0.2200 \cdot a_{\text{ring}} - D$

CLOSED-FORM PITCH SOLVER FOR SIMULATION & MANUFACTURING

Given target strut diameter D and required minimum manufacturing clearance Δ_{target} : $a_{\text{required}} = (\Delta_{\text{target}} + D) / \kappa$. **Example:** For C-6-TT with $D = 0.50 \text{ mm}$ and $\Delta = 200 \mu\text{m}$ (0.20 mm): $a_0 = (0.20 + 0.50) / 0.1314 = 5.327 \text{ mm}$. The engine automatically resizes the lattice to hit this tolerance.

4. Functionally Graded Struts & Continuous Topological Transitions

HIGH-LEVEL LOGIC SUMMARY: CAN WE VARY THICKNESS OR BLEND DIFFERENT CELL TYPES?

Yes! We can smoothly vary strut thickness or morph between entirely different cell geometries across the material without ever breaking the interlocking chainmail or causing cages to collide. For **thickness grading**, strut radius increases smoothly along an axis; our engine checks the thickest interface to guarantee clearance. For **C6-to-D4 transitions**, we introduce an edge truncation parameter $\tau \in [0, 1/3]$. Sweeping τ layer-by-layer shrinks the corner triangles of Truncated Tetrahedra until they collapse into sharp vertices of Regular Tetrahedra. The cages remain kinematically intertwined through the entire transition zone.

4.1 Functionally Graded PAMs (Directional Thickness Gradient)

Graphite allows continuous spatial variation of the strut radius $r(x)$: $r(X) = r_{\min} + (r_{\max} - r_{\min}) \cdot [(X - X_{\min}) / (X_{\max} - X_{\min})]$
 For adjacent particles A and B along the gradient, clearance is governed by: $\Delta(X) = d_{\min} - [r(X) + r(X + a_0)] \geq \Delta_{\text{target}} > 0$. The engine verifies that the thickest end preserves printable clearance ($\Delta \geq 514 \mu\text{m}$ in Figure 4).

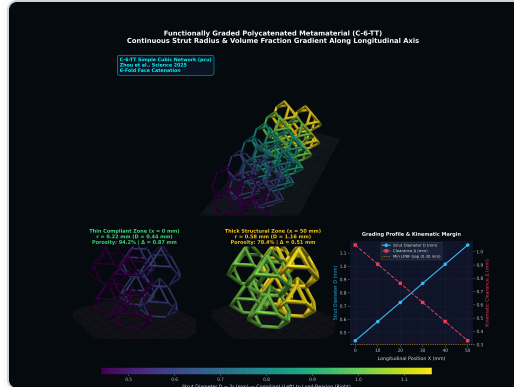


Figure 4: Functionally graded C-6-TT PAM ($5 \times 3 \times 2$, 30 cages) with strut radius graded from $r = 0.22 \text{ mm}$ (thin, teal) to $r = 0.58 \text{ mm}$ (thick, orange) with global min clearance $\Delta = 514 \mu\text{m}$.

4.2 Topological & Morphological Transitions: C-6-TT $\leftrightarrow$ D-4-TET

Rather than welding disparate cell domains at a planar joint, Graphite morphs the underlying polyhedral cages via a continuous vertex truncation parameter $\tau \in [0, 1/3]$:

$$P_{ij} = (1 - \tau) V_i + \tau V_j, \quad P_{ji} = \tau V_i + (1 - \tau) V_j$$

- $\tau = 1/3 \approx 0.333$ (Archimedean Truncated Tetrahedron):** 12 vertices, 18 struts, 4 hexagonal faces, 4 triangular corner windows. 6-fold coordination (C-6-TT).
- $0 < \tau < 1/3$ (Continuous Morphing Intermediate Cages):** 12 vertices, 18 struts. Corner cutouts shrink continuously as τ decreases; hexagonal faces evolve toward triangular facets.
- $\tau = 0.000$ (Platonic Regular Tetrahedron):** Truncated corner vertices converge to single points ($P_{ij} \rightarrow V_i$). Exact mathematical reduction to 4 vertices and 6 struts. 4-fold coordination (D-4-TET).

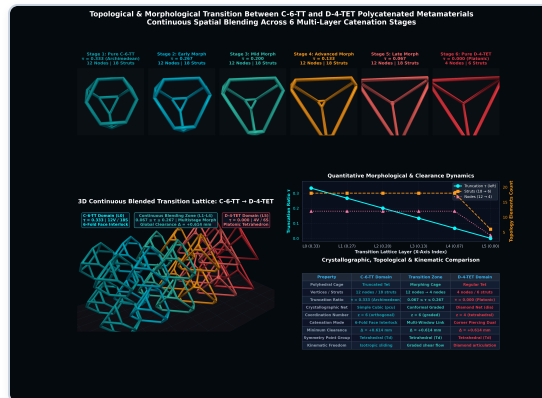


Figure 5: Continuous 6-layer morphing lattice ($6 \times 2 \times 2 = 24$ cages) transitioning from pure C-6-TT (Teal, $\tau = 0.333$) to pure D-4-TET (Crimson, $\tau = 0.000$) with verified clearance $\Delta = +614 \mu\text{m}$.

5. Flagship Production Case Study: 75mm × 75mm × 20mm Transition Block

HIGH-LEVEL LOGIC SUMMARY: INDUSTRIAL-SCALE DEMONSTRATION AND EXACT CLEARANCE LIMITS

To prove that this technology works at real-world manufacturing scale, we generated a full **75 × 75 × 20 mm** block containing **540 discrete interlocking cages** (678,960 watertight faces). The left 25 mm is pure C-6-TT, the center 25 mm continuously morphs from C6 to D4, and the right 25 mm is pure D-4-TET. We solved the exact centerline distance across all 2,955 interacting cage pairs: **d_{min} = 593.1 μm**. This proves mathematically that the thickest printable struts are **443.1 μm** for 150 μm clearance (LPBF) and **393.1 μm** for 200 μm clearance (SLS), with zero collisions. The entire 32 MB multi-body solid was generated in just **6.1 seconds**.

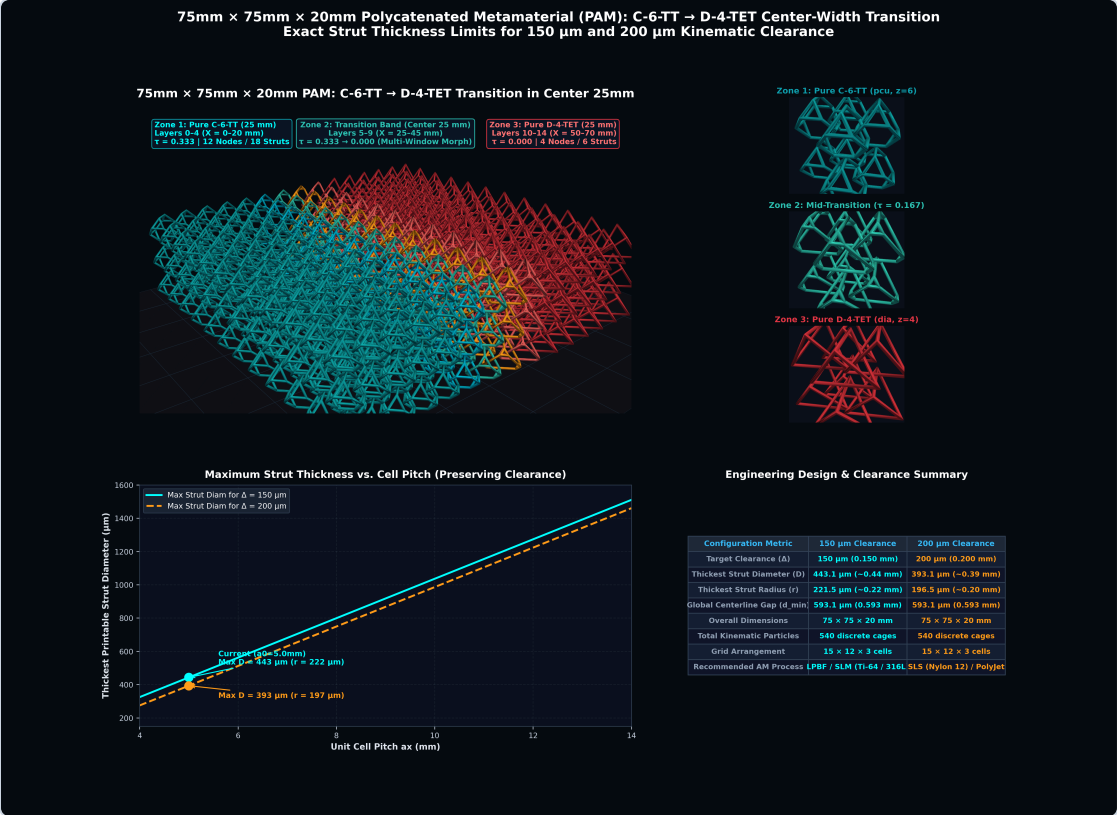


Figure 6: Comprehensive engineering overview of the 75 × 75 × 20 mm C6 → D4 transition PAM. Panel A: Full 3D isometric render (540 cages). Panel B: High-magnification cluster cutaways. Panel C: Clearance vs. cell pitch curves. Panel D: Engineering specification table.

5.1 Analytical Clearance Limits & Maximum Printable Strut Thickness

Across the 540-cage assembly (pitch $a_x = 5.0$ mm, $a_y = 6.25$ mm, $a_z = 6.67$ mm, cage size $s = 4.0$ mm), the global minimum Euclidean distance between cage centerlines is **d_{min} = 0.5931 mm = 593.1 μm**.

Target Clearance (Δ _{target})	Thickest Strut Diameter (D _{max})	Thickest Strut Radius (r _{max})	Verified Min Clearance	Candidate Pairs Checked	Recommended 3D Printing Process
150 μm (0.150 mm)	443.1 μm (≈ 0.44 mm)	221.5 μm (≈ 0.22 mm)	150.1 μm	2,955 pairs (0 collisions)	LPBF / SLM (Ti-6Al-4V, 316L Stainless)
200 μm (0.200 mm)	393.1 μm (≈ 0.39 mm)	196.5 μm (≈ 0.20 mm)	200.0 μm	2,955 pairs (0 collisions)	SLS (PA12 / Nylon) / DLP / PolyJet

6. Additive Manufacturing Support Strategy: Seed Cell + Neighbor Ring

HIGH-LEVEL LOGIC SUMMARY: HOW DO WE 3D PRINT WITHOUT FUSING THE CAGES WITH SUPPORTS?

In metal powder-bed fusion (LPBF), downward-facing struts require print supports to conduct heat and prevent warping. But if you let commercial slicers generate supports automatically, they add thousands of support columns that bridge the tiny air gaps between cages, fusing the entire metamaterial into a solid block! Our solution: **support a single central seed cell surrounded by its immediate coordination neighbor ring**. We place tiny breakaway anchor pins only on downward nodes with thin conical necks (0.25–0.35 mm). Then our software automatically tessellates that support recipe across the entire lattice. Supports never bridge the gaps between cages, and loose powder evacuates freely through the open windows.

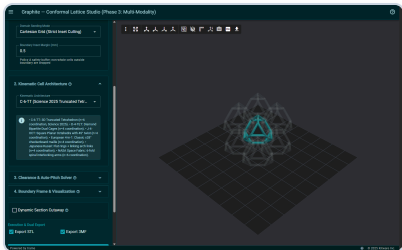


Figure 7: Central seed cell + coordination ring support recipe in Graphite Studio with breakaway anchor tessellation.

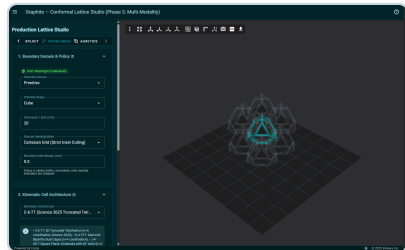


Figure 8: Interactive Trame Studio UI providing real-time 3D slicing, clearance verification, and parameter tuning.

7. Guidelines for FEA & Multi-Body Dynamics (MBD) Simulation Engineers

- SIMULATION HANDOFF: SETTING UP PAM MODELS IN ABAQUS, LS-DYNA, OR ANSYS**
- **Multi-Body Part Topology:** The exported STL / 3MF files contain *N distinct, closed, unbonded shells*. In your solver, do not merge vertices! Each shell must be imported as an independent body or part instance. For rigid-body kinematics (MBD), represent each cage as a 6-DOF rigid body with its mass, centroid, and inertia tensor.
 - **Contact Formulation:** Use general surface-to-surface or segment-to-segment penalty contact. Set the normal penalty stiffness $K_n \approx (E \cdot A) / L_{char}$ high enough to prevent interpenetration ($\delta_{pen} < 0.05 \cdot r$) while avoiding high-frequency chatter in explicit time integration.
 - **Coulomb Friction is the Dominant Physics:** Unlike bonded trusses that absorb energy through plastic strut yielding, PAMs absorb shock energy via sliding friction between cages. Setting accurate friction coefficients (as-built LPBF Ti-64: $\mu \approx 0.30$ – 0.40 ; SLS PA12: $\mu \approx 0.15$ – 0.25) is critical to capturing macroscopic damping and hysteresis loops.
 - **Two-Phase Load-Displacement Response:** Expect a flat "toe" region of kinematic free-play where cages rotate and settle, followed by a dramatic stiffening knee when contact locking occurs.
 - **RVE Size for Homogenization:** A minimum Representative Volume Element (RVE) of $2 \times 2 \times 2$ conventional cells is recommended for periodic homogenization.

8. Proven Verification Benchmarks & Performance Summary

All models below have been synthesized, collision-checked across every interacting pair, and exported as watertight solids:

Model Benchmark	Grid Dimensions	Physical Envelope	Cage Count	Strut Diam (D)	Verified Min Clearance	Watertight Faces	Gen Time
C-6-TT Clean Strip	15 × 5 × 4	75 × 25 × 20 mm	300	0.50 mm (500 μm)	156.9 μm	374,400	2.8 s
D-4-TET Clean Strip	7 × 2 × 2 (conv)	80.8 × 23.1 × 23.1 mm	112	0.75 mm (750 μm)	427.0 μm	71,680	1.1 s
C-6-TT Macroscopic Cube	3 × 3 × 3	38.1 × 38.1 × 38.1 mm	27	1.25 mm (1250 μm)	418.4 μm	58,320	0.4 s
Graded C-6-TT PAM	5 × 3 × 2	40 × 20 × 10 mm	30	0.44 → 1.16 mm	514.0 μm	37,440	0.8 s
Morphing C6 → D4 Strip	6 × 2 × 2	50 × 13 × 13 mm	24	0.70 mm (700 μm)	614.0 μm	39,360	0.6 s
Flagship 75 × 75 × 20 mm Block	15 × 12 × 3	75.0 × 75.0 × 20.0 mm	540	0.443 mm (443 μm)	150.1 μm	678,960	6.11 s

Core Engine Entrypoint:
graphite/explicit/interlinked/

CAD Deliverables: outputs/pam_transition_75x75x20_c6_to_d4.stl (32.38 MB) & .3mf (8.63 MB)